

TORSION IN GRAPH MAGNITUDE HOMOLOGY

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Preliminary Examination

Outline of presentation:

- Part I:
 - ① Magnitude $\#G$
 - ② Magnitude homology $MH_{k,\ell}(G)$
 - ③ Computational tools: PIE & MV
 - ④ Diagonality in magnitude homology
- Part II:
 - ① Posets & KY-construction
 - ② Extending KY
 - ③ Our results
 - ④ Eulerian magnitude homology

MAGNITUDE OF METRIC SPACES

DEFINITION (LEINSTER '13 [12])

The **similarity matrix** of a finite metric space (X, d) is

$$\zeta_X := \begin{pmatrix} q^{d(x_1, x_2)} & \dots & q^{d(x_1, x_n)} \\ \vdots & \ddots & \vdots \\ q^{d(x_n, x_1)} & \dots & q^{d(x_n, x_n)} \end{pmatrix}$$

where $X = \{x_1, \dots, x_n\}$. If ζ_X is invertible, then X has **Möbius inversion** with **Möbius matrix** $\mu_X = \zeta_X^{-1}$.

The **magnitude** $\#X(q)$ of X is the sum of the entries in μ_X .

REMARK

Graphs are metric spaces when equipped with the shortest-path distance $d : V(G) \times V(G) \rightarrow \mathbb{Z}$ between vertices.

MAGNITUDE OF A GRAPH

DEFINITION (LEINSTER '14 [11])

The **magnitude** $\#G(q)$ of a finite simple connected graph $G = (V(G), E(G))$ is the sum of the entries of $\mu_G = \zeta_G^{-1}$

$$\#G(q) := \sum_{a,b \in V(G)} \mu_G(a, b).$$

- $\zeta_G(0) = I$ leinster [11].
- $\det(\zeta_G(q))$ is nonzero and thus invertible in $\mathbb{Q}(q)$ [11].
- G always has Möbius inversion [11].

REMARK

We may at times write $\#G$ instead of $\#G(q)$ for convenience.

MAGNTIUDE AS A WEIGHTING FUNCTION

DEFINITION (LEINSTER '08 [10])

A **weighting** on ζ_X is a function $w : X \rightarrow \mathbb{Q}(q)$ such that

$$\sum_{b \in X} \zeta(a, b) w(b) = 1$$

for all $a \in X$.

REMARK

If ζ_X is invertible, then the weighting on ζ_X is uniquely determined by μ_X .

COROLLARY

Given a weighting on an invertible ζ_X ,

$$\#X(q) = \sum_{b \in X} w(b).$$

A GENERAL FORMULA FOR MAGNITUDE

PROPOSITION (LEINSTER '14 [11])

For any graph G

$$\#G(q) = \sum_{k \geq 0} (-1)^k \sum_{\substack{x_i \neq x_{i+1} \\ 0 \leq i \leq k}} q^{d(x_0, x_1) + d(x_1, x_2) + \cdots + d(x_{k-1}, x_k)}$$

where each $x_i \in V(G)$.

Let $y \in V(G)$, and define

$$w(y) = \sum_{k \geq 0} (-1)^k \sum_{\substack{y \neq y_1 \\ y_i \neq y_{i+1} \\ 1 \leq i < k}} q^{d(y, y_1) + \cdots + d(y_{k-1}, y_k)}.$$

A GENERAL FORMULA FOR MAGNITUDE

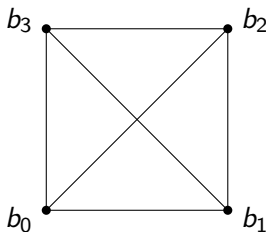
$$\begin{aligned}\sum_{y \in V(G)} q^{d(x,y)} w(y) &= w(x) + \sum_{\substack{y \in V(G) \\ y \neq x}} q^{d(x,y)} w(y) \\ &= w(x) + \sum_{k \geq 0} (-1)^k \sum_{\substack{x \neq y \\ y_i \neq y_{i+1} \\ 1 \leq i < k}} q^{d(x,y) + d(y,y_1) + \dots + d(y_{k-1}, y_k)} \\ &= w(x) + \sum_{k \geq 0} (-1)^k \sum_{\substack{x \neq x_1 \\ x_i \neq x_{i+1} \\ 1 \leq i < k+1}} q^{d(x,x_1) + \dots + d(x_k, x_{k+1})} \\ &= 1.\end{aligned}$$

CATEGORIFYING THE MAGNITUDE OF A GRAPH

DEFINITION (HEPWORTH–WILLERTON '15 [6])

- A ***k-path*** is a $(k + 1)$ -tuple of vertices of G , $v = (v_0, \dots, v_k)$, such that $v_i \neq v_{i+1}$.
- The ***length of a k-path*** is $\text{len}(v) = \sum_{i=0}^{k-1} d(v_i, v_{i+1})$.
- The set of all k -paths in G of length ℓ is $T_{k,\ell}(G)$.
- The (k, ℓ) -***magnitude chain group*** is $MC_{k,\ell}(G) := \mathbb{Z}\langle T_{k,\ell}(G) \rangle$.

$$MC_{2,2}(G) = \bigoplus_{h \neq i \neq j} \mathbb{Z}\langle (b_h, b_i, b_j) \rangle$$



CATEGORIFYING THE MAGNITUDE OF A GRAPH

DEFINITION (HEPWORTH–WILLERTON '15[6])

- The boundary is $\partial_{k,\ell} : MC_{k,\ell}(G) \rightarrow MC_{k-1,\ell}(G)$ where $\partial_{k,\ell} = \sum_{i=1}^{k-1} (-1)^i \partial_{k,\ell}^i$ and

$$\partial_{k,\ell}^i(v) = \begin{cases} v^{\setminus i} = (v_0, \dots, \widehat{v_i}, \dots, v_k) & \text{len}(v^{\setminus i}) = \ell \\ 0 & \text{otherwise.} \end{cases}$$

- The (k, ℓ) -**magnitude homology group** $MH_{k,\ell}(G)$ is the homology of the magnitude chain complex.

OBSERVATION (ASAO–IZUMIHARA '20 [2])

The magnitude chain complex of a graph breaks down into a direct sum decomposition which the homology inherits

$$MC_{k,\ell}(G) = \bigoplus_{a,b \in V(G)} MC_{k,\ell}(a, b).$$

CATEGORIFYING THE MAGNITUDE OF A GRAPH

THEOREM (HEPWORTH–WILLERTON '15 [6])

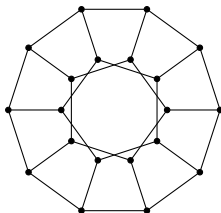
Magnitude homology categorifies magnitude of a graph:

$$\sum_{k,\ell \geq 0} (-1)^k \operatorname{rk}(MH_{k,\ell}(G)) q^\ell = \#G.$$

$$\begin{aligned} \sum_{\ell \geq 0} \chi(MH_{*,\ell}(G)) q^\ell &= \sum_{\ell \geq 0} \chi(MC_{*,\ell}(G)) q^\ell \\ &= \sum_{\ell \geq 0} \sum_{k \geq 0} (-1)^k \operatorname{rk}(MC_{k,\ell}(G)) q^\ell \\ &= \sum_{k \geq 0} (-1)^k \sum_{\ell \geq 0} \operatorname{rk}(MC_{k,\ell}(G)) q^\ell \\ &= \sum_{k \geq 0} (-1)^k \sum_{\substack{x_i \neq x_{i+1} \\ 0 \leq i < k}} q^{d(x_0, x_1) \cdots + d(x_{k-1}, x_k)} \end{aligned}$$

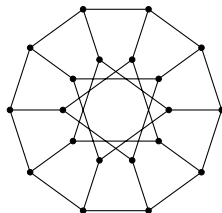
MAGNITUDE HOMOLOGY IS A STRONGER INVARIANT

- Gu '18 [5] Examples of graphs with equal magnitude but different magnitude homology



Dodecahedral graph DC

$$MH_{2,4}(DC) = \mathbb{Z}^{60}$$



Deasargues graph DS

$$MH_{2,4}(DS) = 0$$

$$\#DC = \#DS = \frac{20}{1 + 3q + 6q^2 + 6q^3 + 3q^4 + q^5}$$

AN INCLUSION-EXCLUSION FORMULA FOR MAGNITUDE

DEFINITION

- A subgraph U of G is **convex in G** if $d|_U(a, b) = d(a, b)$ for $a, b \in U$.
- Let $V_U(G)$ consist of all vertices in G in the same connected component as U .
- G **projects to U** if for all $g \in V_U(G)$ there exists a unique $\pi(g) \in V(U)$ such that for all $u \in U$

$$d(g, u) = d(g, \pi(g)) + d(\pi(g), u).$$

LEMMA (INCLUSION-EXCLUSION, LEINSTER '14 [11])

For a graph $G = A \cup B$. If $A \cap B$ is convex in G and either A or B projects to $A \cap B$, then

$$\#G = \#A + \#B - \#(A \cap B)$$

A MAYER-VIETORIS SEQUENCE FOR MAGNITUDE HOMOLOGY

THEOREM (MAYER-VIETORIS SEQUENCE, HEPWORTH–WILLERTON '15[6])

For a graph $G = A \cup B$, if $A \cap B$ is convex in G and either A or B projects to $A \cap B$, there is a short exact sequence

$$0 \rightarrow MH_{k,\ell}(A \cap B) \rightarrow MH_{k,\ell}(A) \oplus MH_{k,\ell}(B) \rightarrow MH_{k,\ell}(G) \rightarrow 0.$$

COROLLARY (HEPWORTH–WILLERTON '15[6])

The Mayer-Vietoris sequence for magnitude homology descends to the Inclusion-Exclusion formula for magnitude.

COROLLARY (HEPWORTH–WILLERTON '15[6])

The magnitude homology groups of trees are torsion-free and all lie on the diagonal $k = \ell$.

DIAGONAL GRAPHS

DEFINITION

A graph G is **diagonal** if $MH_{k,\ell}(G) \neq 0$ only when $k = \ell$.

- Trees are diagonal (Hepworth-Willerton '15)
- Pawful graphs are diagonal (Gu '18)
- The icosahedral graph is diagonal but not pawful (Gu '18)
- Diagonal graphs always have torsion-free magnitude homology (Hepworth-Willerton '17)
- For any two graphs with nonempty vertex set their join is diagonal (Hepworth-Willerton '17)

PROPOSITION (HEPWORTH-WILLERTON '15[6])

If G is diagonal, then the magnitude $\#G$ of G determines $MH_{*,*}(G)$.

IMPLICATIONS OF DIAGONALITY

THEOREM (SAZDANOVIC–SUMMERS '19)

If a graph G does not contain C_3 or C_4 as a subgraph, we have that $MH_{0,0}(G) \cong \mathbb{Z}\langle V(G) \rangle$ and $MH_{k,k}(G) \cong \mathbb{Z}\langle E(G) \times \{0,1\} \rangle$ for $k > 0$.

DEFINITION

*The **local girth** of G at $x \in V(G)$ is the length of the smallest cycle containing x . The **girth** $\text{gir}(G)$ of G is the infimum of local girths over all vertices of G . If G is a tree then $\text{gir}(G) = \infty$.*

THEOREM (ASAO–HIRAOKA–KANAZAWA '21)

If G is diagonal then $\text{gir}(G) = 3, 4$ or ∞

THE STORY SO FAR

- Leinster came up with a new invariant for metric spaces called magnitude.
- Graphs are metric spaces so magnitude is also an invariant for graphs.
- Magnitude was categorified into magnitude homology.
- Magnitude satisfies an inclusion-exclusion formula which lifts to a Mayer-Vietoris sequence.
- Graphs that have diagonal magnitude homology have restrictions on their girth.
- But what do we know about the appearance of torsion in magnitude homology?

POSETS AND MAGNITUDE HOMOLOGY

DEFINITION

For a poset P let $G(P)$ be the graph representing the Hasse diagram of $\hat{P}$. If P is the face poset of a simplicial complex S , we write $G(S)$ instead.

DEFINITION

The order complex ΔP of a poset P is the complex with chain groups generated by chains in a poset

$$C_k(\Delta P) := \mathbb{Z}\langle (p_0 < p_1 < \cdots < p_k) : p_i \in P \rangle$$

PROPOSITION (KANETA–YOSHINAGA '18[8])

If P is a ranked poset, there is a natural embedding

$$H_k(\Delta P) \hookrightarrow MH_{k+2, rk(\hat{P})}(G(P)).$$

THE KANETA-YOSHINAGA CONSTRUCTION

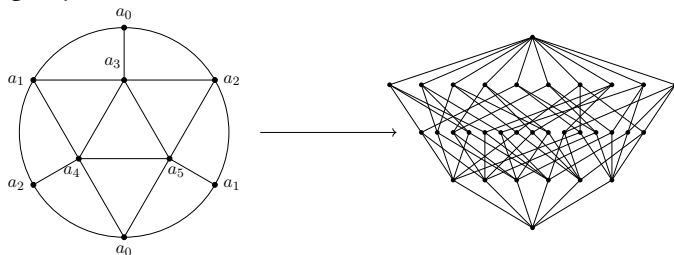
THEOREM (KANETA-YOSHINAGA '18 [8])

Given a triangulable manifold K , let F be the face poset of K . Using the Kaneta-Yoshinaga embedding, we have

$$H_k(K) \cong H_k(Bd(K)) \cong H_k(\Delta F) \hookrightarrow MH_{k+2, rk(\hat{F})}(G(K))$$

COROLLARY

The graph $G(\mathbb{RP}^2)$ contains $\mathbb{Z}_2$ as a subgroup in its $(3, 4)$ -magnitude homology group.



TORSION IN MAGNITUDE HOMOLOGY

THEOREM (SAZDANOVIC–SUMMERS '19 [15])

For any finitely generated abelian group

$$A = \mathbb{Z}^r \oplus \mathbb{Z}_{p_1^{r_1}} \oplus \cdots \oplus \mathbb{Z}_{p_m^{r_m}}$$

there exists a graph G and integers $k, \ell \geq 0$ such that $A \trianglelefteq MH_{k,\ell}(G)$.

- The lens space $L(p^r, q_1, q_2, \dots, q_n)$ is a triangulable $(2n + 1)$ -dim manifold with fundamental group isomorphic to $\mathbb{Z}_{p^r}$
- Use connect sums of lens spaces with solid torii and the Kaneta Yoshinaga construction.
- Magnitude homology of $G(K)$ obtained from a triangulation K of such a space contains A .

THEOREM (SAZDANOVIC–SUMMERS '19 [15])

There exist graphs with $\mathbb{Z}_{p^r} \trianglelefteq MH_{3,2n+3}(G)$ for prime p .

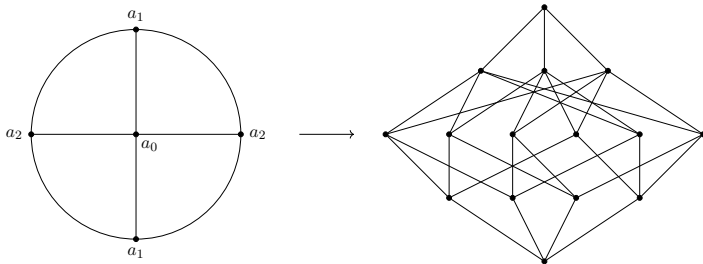
SAY “NO!” TO TRIANGULABLE MANIFOLDS

DEFINITION

For a regular CW-complex X , define the face poset of X to be the poset obtained through nonempty cell closure inclusion.

DEFINITION

For a regular CW-complex X and its face poset F , let $G(X)$ be the graph representing the Hasse diagram of $\hat{F}$.



WHY REGULAR CW-STRUCTURES?

DEFINITION (KOZLOV [9])

The **barycentric subdivision** $Bd X$ of a regular CW complex X is a simplicial complex which has as k -simplices exactly those sequences $F_1 \subset F_2 \subset \cdots \subset F_k$ of closures of cells F_i of X .

PROPOSITION (KOZLOV [9])

For a regular CW complex X we have that

$$Bd X = \Delta F$$

where F is the face poset of X . Furthermore,

$$|Bd X| \approx |\Delta F| \approx |X|$$

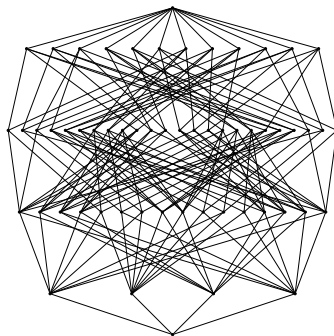
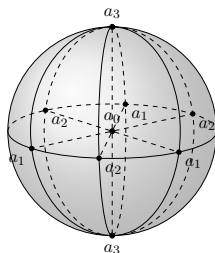
where $|\cdot|$ represents the geometric realization and $\approx$ denotes homeomorphism.

EXTENDING THE KANETA–YOSHINAGA CONSTRUCTION

THEOREM (M.–SAZDANOVIC '25[7])

Given a topological space X with a regular n -dimensional CW-structure, there exists an embedding of $H_k(X)$ into $MH_{k+2,n+2}(G(X))$.

The graph $G(L(3,1))$ contains $\mathbb{Z}_3$ as a subgroup in its $(3,5)$ -magnitude homology group.



PROPOSITION (M.–SAZDANOVIC '25[7])

Every Moore space $M(\mathbb{Z}_{p^r}, k)$ can be equipped with a regular CW-structure.

Proof sketch: by induction

- Begin with $M(\mathbb{Z}_{p^r}, 1)$
- Assume induction hypothesis for $M(\mathbb{Z}_{p^r}, n)$ and extend to $M(\mathbb{Z}_{p^r}, n + 1)$ by embedding the regular CW structure of $M(\mathbb{Z}_{p^r}, n)$ into the hemisphere of D^{n+2} .
- Add north and south poles and identify the $n + 1$ dimensional faces on the boundary of D^{n+2} appropriately.
- By construction, the boundary of each cell is homeomorphic to a sphere.

EXTENDING THE SAZDANOVIC–SUMMERS THEOREM

COROLLARY (M.–SAZDANOVIC '25[7])

Given a finitely generated abelian group A and an integer $k > 2$, there exists a graph G with $A \trianglelefteq MH_{k,k+1}(G)$.

PROOF.

Using wedge sums of Moore spaces and balls without origin we get

$$\begin{aligned} A &\cong \mathbb{Z}_{p_1^{r_1}} \oplus \mathbb{Z}_{p_2^{r_2}} \oplus \cdots \oplus \mathbb{Z}_{p_n^{r_n}} \oplus \mathbb{Z}^m \\ &\cong H_{k-2} \left(M(\mathbb{Z}_{p_1^{r_1}}, k-2) \vee \cdots \vee M(\mathbb{Z}_{p_n^{r_n}}, k-2) \vee \bigvee_{i=1}^m \mathbb{D}^{n-1} \setminus \{0\} \right) \end{aligned}$$

which we can embed into the graph of the $\widehat{F}$ of the regular CW-structure associated to this space. □

THE ASAO-IZUMIHARA COMPLEX

DEFINITION (ASAO-IZUMIHARA '20[2])

For a graph G and a pair of vertices $a, b \in V(G)$, define the **Asao-Izumihara complex** of G to be

$$K_\ell(a, b) = \{ \{ (x_{i_1}, i_1), \dots, (x_{i_k}, i_k) \} \mid k \leq \ell - 1, \text{len}(a, x_{i_1}, \dots, x_{i_k}, b) \leq \ell \}$$

$$K'_\ell(a, b) = \{ \{ (x_{i_1}, i_1), \dots, (x_{i_k}, i_k) \} \mid k \leq \ell - 1, \text{len}(a, x_{i_1}, \dots, x_{i_k}, b) < \ell \}.$$

THEOREM (ASAO-IZUMIHARA '20[2])

For the following relative chain groups,

$$C_k(K_\ell(a, b), K'_\ell(a, b)) \cong MC_{k+2, \ell}(a, b).$$

COROLLARY (ASAO-IZUMIHARA '20[2])

$$H_k(K_\ell(a, b), K'_\ell(a, b)) \cong MH_{k+2, \ell}(a, b).$$

LOWER BOUNDS FOR THE APPEARANCE OF TORSION

COROLLARY (ASAO–ISUMIHARA '20[2])

Given a graph G and vertices $a, b \in V(G)$

$$MH_{2,\ell}(a, b) \cong \begin{cases} H_0(K_\ell(a, b), K'_\ell(a, b)) & \text{if } d(a, b) < \ell \\ \tilde{H}_0(K_\ell(a, b)) & \text{if } d(a, b) = \ell \end{cases}$$

COROLLARY (M.–SAZDANOVIC '25[7])

For a graph G all magnitude homology groups $MH_{2,\ell}(G)$ must be torsion-free.

REMARK

Thus for torsion to appear, it must be in bigradings with $\ell > k > 2$.

A MAYER-VIETORIS SEQUENCE FOR POSETS

DEFINITION (PRETZEL '69[14])

For a poset P and any two subsets $A, B \subseteq P$ define

$$A_B := \{a \in A : a \leq b \text{ or } b \leq a, \forall b \in B\}.$$

THEOREM (PRETZEL '69[14])

Given a poset $P = A \cup B$, if any finite totally ordered subset $M \subseteq P$ satisfies the following:

- $M \subseteq A$
- $M \subseteq B$
- $\tilde{H}_k((A \cap B)_M) = 0$ for all k

then the following sequence is exact

$$\dots \rightarrow \tilde{H}_k(A \cap B) \rightarrow \tilde{H}_k(A) \oplus \tilde{H}_k(B) \rightarrow \tilde{H}_k(P) \rightarrow \tilde{H}_{k-1}(A \cap B) \rightarrow \dots$$

COMPLETELY DESCRIBING THE SUPPORT OF TORSION

THEOREM (M.–SAZDANOVIC '25[7])

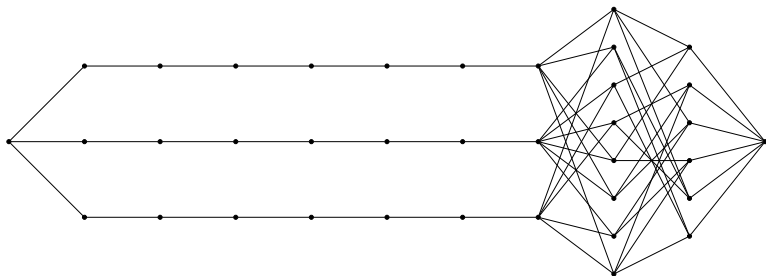
Given any integers $\ell > k > 2$ and a finitely generated abelian group A there exists a graph G such that $A \trianglelefteq MH_{k,\ell}(G)$.

Proof sketch:

- Begin with the bigrading $(k, k + 1)$ using the wedge sum of Moore spaces and spheres.
- Count the atoms of the face poset F of the regular CW-structure associated to the space.
- Given $\ell > k + 1$, let $m = \ell - (k + 1)$ and wedge a disjoint union of totally ordered posets of length m to the atoms.
- Use Mayer-Vietoris for posets to get $A \trianglelefteq \tilde{H}_{k-2}(\Delta F)$.

PATRICK'S 3-TORSION SWORD

The following graph contains $\mathbb{Z}_3$ in its $(3, 10)$ -magnitude homology



EULERIAN MAGNITUDE HOMOLOGY

DEFINITION (GIUSTI–MENARA '24[4])

- An **Eulerian k -path** is a $(k + 1)$ -tuple of vertices of G , $v = (v_0, \dots, v_k)$, such that $v_i \neq v_j$ whenever $i \neq j$.
- The set of all Eulerian k -paths in G of length ℓ is $ET_{k,\ell}(G)$.
- The (k, ℓ) -**Eulerian magnitude chain group** is $EMC_{k,\ell}(G) := \mathbb{Z}\langle ET_{k,\ell}(G) \rangle$.
- The (k, ℓ) -**Eulerian magnitude homology group** is

$$EMH_{k,\ell}(G) = \frac{\ker \partial_{k,\ell}|_{EMC_{k,\ell}(G)}}{\operatorname{Im} \partial_{k+1,\ell}|_{EMC_{k,\ell}(G)}}$$

- The (k, ℓ) -**discriminant magnitude chain group** is $DMC_{k,\ell}(G) = MC_{k,\ell}(G)/EMC_{k,\ell}(G)$ with (k, ℓ) -**discriminant magnitude homology** $DMH_{k,\ell}(G)$.

EULERIAN MAGNITUDE HOMOLOGY

OBSERVATION (MENARA 24'[13])

Eulerian magnitude homology also shares the direct sum decomposition

$$EMH_{k,\ell}(G) \cong \bigoplus_{a \neq b \in V(G)} EMH_{k,\ell}(a, b)$$

THEOREM (GIUSTI-MENARA '24[4])

These three magnitude homologies fit into a long exact sequence

$$\dots \rightarrow EMH_{k,\ell}(G) \rightarrow MH_{k,\ell}(G) \rightarrow DMH_{k,\ell}(G) \rightarrow EMH_{k-1,\ell}(G) \rightarrow \dots$$

COROLLARY (M.-SAZDANOVIC '25[7])

If G is a diagonal graph with $EMH_{k,k}(G) = 0$ and $EMG_{k-1,k}(G)$ torsion-free, then the following sequence splits

$$0 \rightarrow MH_{k,k}(G) \rightarrow DMH_{k,k}(G) \rightarrow EMH_{k-1,k}(G) \rightarrow 0.$$

THE ASAO-ISUMIHARA COMPLEX FOR EMH

DEFINITION (MENARA '24[13])

$ET_\ell(a, b)$ is the subcomplex of $K_\ell(a, b)$ consisting of Eulerian paths.

THEOREM (MENARA '24[13])

For the following relative chain groups,

$$C_k(ET_\ell(a, b), ET_{\ell-1}(a, b)) \cong EMC_{k+2,\ell}(a, b)$$

$$H_k(ET_\ell(a, b), ET_{\ell-1}(a, b)) \cong EMH_{k+2,\ell}(a, b).$$

COROLLARY (M.–SAZDANOVIC '25[7])

Given a graph G and vertices $a, b \in V(G)$

$$EMH_{2,\ell}(a, b) \cong \begin{cases} H_0(ET_\ell(a, b), ET_{\ell-1}(a, b)) & \text{if } d(a, b) < \ell \\ \tilde{H}_0(ET_\ell(a, b)) & \text{if } d(a, b) = \ell \end{cases}$$

THEOREM (M.–SAZDANOVIC '25[7])

Given a ranked poset P we have that

$$MH_{k, rk(\hat{P})}(\hat{0}, \hat{1}) \cong EMH_{k, rk(\hat{P})}(\hat{0}, \hat{1}),$$

for all $k \leq rk(\hat{P})$

Proof sketch: there are two methods of proof

- ① Use the Kaneta-Yoshinaga embedding and the fact that elements in the chain of a poset must all be unique.
- ② Use the Asao-Izumihara complex and observe that

$$K'_{rk(\hat{P})}(\hat{0}, \hat{1}) = \emptyset = ET_{rk(\hat{P})-1}(\hat{0}, \hat{1}).$$

TORSION IN EULERIAN MAGNITUDE HOMOLOGY

CONJECTURE (MENARA '24[13])

For all graphs G and all $\ell \geq k \geq 0$, $EMH_{k,\ell}(G)$ is torsion-free.

COROLLARY (M.–SAZDANOVIC '25[7])

For a graph G all Eulerian magnitude homology groups $EMH_{2,\ell}(G)$ must be torsion-free.

COROLLARY (M.–SAZDANOVIC '25[7])

Given a topological space X with a regular CW-structure, there exists an embedding of $H_k(X)$ into $EMH_{k+2, rk(\widehat{F})}(G(X))$.

COROLLARY (M.–SAZDANOVIC '25[7])

Given any two integers $\ell > k > 2$ and a finitely generated abelian group A there exists a graph G with $A \trianglelefteq EMH_{k,\ell}(G)$.

FUTURE DIRECTIONS

- **DMH-only torsion challenge.** Construct graphs G with torsion in $DMH_{*,*}(G)$ but not in $MH_{*,*}(G)$ nor $EMH_{*,*}(G)$.
- **Hypergraph magnitude homology.** The magnitude homology of a hypergraph is the magnitude homology of its weighted incidence graph, relating it to ordinary magnitude homology of graphs. Can this be exploited to develop computational tools for hypergraph magnitude homology[3]?
- **Poset torsion.** Poset structure encoded by the torsion in its order complex homology may yield some insight on what graph structure is encoded by torsion in graph magnitude homology.
- **Random graphs and windows for torsion.** Use random graphs $G(n, n^{-\alpha})$ to study the behavior of torsion in magnitude homology theories as $n \rightarrow \infty$ [13][1].

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